# System Architecture

SmartFall: Watch-based Application


Architecture Overview:..... 1
Configuration:..... 2
Example Setup (Default):..... 3


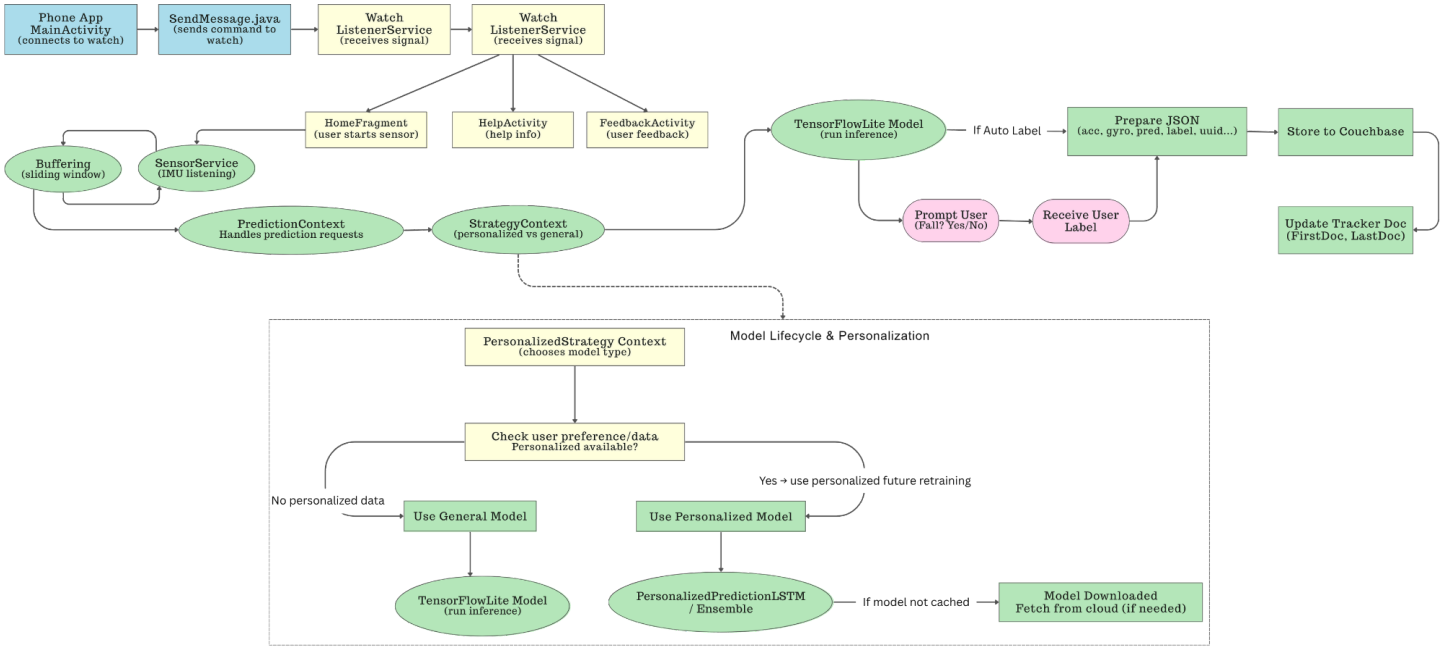


Figure 01: Flowchart of the system architecture

## Architecture Overview:

As illustrated in Figure 01, the **SmartFall system** follows a modular, event-driven architecture that integrates mobile-wearable communication, real-time sensor processing, machine learning inference, and cloud-synced data storage.

### Phone-to-Watch Activation

- The **Phone App** acts as a lightweight controller. It initiates communication with the **Watch App** using SendMessage.java.
- The **Watch ListenerService** receives the signal and launches the main activity, initializing the user interface and background services.

### Sensor Data Collection and UI Flow

- From the **HomeFragment**, the user can start or stop IMU data collection.
- The **SensorService** continuously listens to accelerometer and gyroscope streams.
- Data is buffered into sliding windows (e.g., 128-sample segments) for inference.

### Model Prediction Pipeline

- The buffered IMU window is passed to PredictionContext.java, which routes the data through the appropriate model.
- StrategyContext determines whether to use a **general** or **personalized** model based on configuration and user availability.
- The selected model (e.g., TensorFlowLiteLSTM or PersonalizedPredictionLSTM) performs fall classification on-device.

### User Label Prompting

- If the prediction confidence is below threshold or manual confirmation is enabled, a **prompt is shown to the user** (e.g., “Was this a fall?”).
- The user's input is captured and used to **label the data** for storage and future personalization.

### Data Packaging and Storage

- The result (sensor data + prediction + label + metadata) is formatted as a JSON object.

```
{
"uuid": "user123",
"timestamp": "2025-06-20T14:33:00Z",
"acc": [...],
"gyro": [...],
"prediction_label": "fall",
"confidence": 0.91,
"model_used": "PERSONALIZED_LSTM",
"type": "FP"
}
```

- The labeled instance is stored in **Couchbase**, both locally and optionally synced to a remote bucket.
- A **tracker document** is updated to maintain session boundaries (e.g., firstDoc, lastDoc).

### Personalization Lifecycle

- Labeled data can be used in future sessions to fine-tune or switch to personalized models.
- The model downloader fetches updated model weights if not already cached on-device.

## Configuration:

The **Watch app** is fully configurable via the **SmartFallConfig.java** file (*wear > java > config > SmartFallConfig.java*). It controls model selection, personalization strategy, runtime behaviour, and remote database settings.

### Configuration Parameters:

| Parameter | Possible Values | Description |
|---|---|---|
| MODEL_TYPE | "PERSONALIZED", "LSTM", "DEFAULT" | Determines the overall model strategy:<br>• PERSONALIZED uses per-user adaptation<br>• LSTM uses general pre-trained model<br>• DEFAULT (fallback/backup logic) |
| PERSONALIZED_STRATEGY | "LSTM", "ENSEMBLE" | Specifies which personalized model architecture to use:<br>• LSTM = single LSTM model<br>• ENSEMBLE = multiple models with majority voting |
| MODEL_FROM | "OFFLINE", "ONLINE" | Determines whether models are loaded:<br>• OFFLINE = from local assets folder<br>• ONLINE = downloaded dynamically |
| OFFLINE_MODEL_FILE | "Saved_model128.tflite" | Name of the local *.tflite* file bundled with the app (used if MODEL_FROM = OFFLINE) |
| OFFLINE_MODEL_THRESHOLD | 0.0–1.0 (e.g., 0.9f) | Confidence threshold below which user confirmation (labelling) is prompted |

| | | |
|---|---|---|
| REMOTE_BUCKET_NAME | "smart-fall-data" | Target Couchbase bucket where labelled documents are uploaded |
| API_KEY | "demoKey!xyz9&'1234ABC" (or others) | Authentication key for syncing with the remote Couchbase Sync Gateway |
| ENV_FLAG | "SMART_FALL", "GANYMEDE" | Environment selector for remote deployment (used to toggle server configs) |
| ALPHA_LIMIT | Integer (e.g., 20) | Length of the **alpha queue** (may represent smoothed output memory or state history) |
| BETA_LIMIT | Integer (e.g., 128) | Size of the input **window** for model inference (i.e., number of IMU samples per prediction) |
| STEP_SIZE | Integer (e.g., 10) | Step size for sliding window advance |

## Example Setup (Default):

```
public static final String MODEL_TYPE = "PERSONALIZED";
public static final String PERSONALIZED_STRATEGY = "LSTM";
public static final String MODEL_FROM = "OFFLINE";
public static final String OFFLINE_MODEL_FILE = "saved_model128.tflite";
public static final float OFFLINE_MODEL_THRESHOLD = 0.9f;
public static final String REMOTE_BUCKET_NAME = "smart-fall-data";
public static final String ENV_FLAG = "SMART_FALL";
public static final int ALPHA_LIMIT = 20;
public static final int BETA_LIMIT = 128;
```